MODULE 1: INTRODUCTION TO PLM AND PDM

1.1 Product Lifecycle Management (PLM)

Product lifecycle management makes it possible to command the whole lifespan of a product and the information connected with it. Efficient product lifecycle management enables companies to compete successfully in international and global markets.

1.1.1 What is a product?

Usually, when talking about products we mean tangible products i.e. goods. The term goods refers to physical, tangible products that can be owned, traded, and distributed to different places at different times without changing their identity. However, a product in a modern world can also be something very intangible such as a piece of software, a piece of knowledge or an algorithm or a formula. They are products as much as tangible products are.

When referring to a product in this book, we refer to three different kinds of products:
1. Goods meaning physical, tangible products
2. Services
3. Intangible products meaning non-physical products that are not services.

For example:

- Software
- An algorithm

1.2 PLM: What is it?

In many ways, product data management can be seen as a subset of PLM. First EDM (Engineering Data emerged in Management) and then PDM (Product Data Management) the late 1980s as engineers in the manufacturing industries recognized a need to keep track of the growing volumes of design files generated by CAD (Computer Aided Design) systems. PDM allowed them to standardize items, to store and control document files, to maintain BOM's, to control item, BOM and document revision levels, and immediately to see relationships between parts and assemblies. This functionality let them quickly access standard items, BOM structures, and files for reuse and derivation, while reducing the risk of using incorrect design versions and increasing the reuse of existing product information.

However, the benefits of operational PLM go far beyond incremental savings, yielding greater bottom-line savings and top-line revenue growth not only by implementing tools and technologies, but also by making necessary, and often tough, changes in processes, practices and methods and gaining control over product lifecycles and lifecycle processes. The return on investment for PLM is based on a broader corporate business value, specifically the greater market share and increased profitability achieved by streamlining the business processes that help deliver innovative, winning products with high brand image quickly to market, while being able to make informed lifecycle decisions over the complete product portfolio during the lifecycle of each individual product. The scope of product information being stored, refined, searched, and shared with PLM has expanded. PLM is a holistic business concept developed to manage a product and its lifecycle including not only items, documents, and BOM's, but also analysis results, test specifications, environmental component information, quality standards, engineering requirements, change orders, manufacturing procedures, product performance information, component suppliers, and so forth. On the other hand, modern PLM system capabilities include workflow, program

management, and project control features that standardize, automate, and speed up product management operations.

Product Lifecycle Management (PLM) is integrated, information driven approach comprised on people, processes/practices, and technology to all aspects of a product's life, from its design through manufacture, deployment and maintenance – culminating in the product's removal from service and final disposal.

Product Lifecycle Management (PLM) is the business activity of managing, in the most effective way, a company's products all the way across their lifecycles; from the very first idea for a product all the way through until it is retired and disposed of.

PLM manages both individual products and the Product Portfolio, the collection of all of a company's products. PLM manages products from the beginning of their life, including development, through growth and maturity, to the end of life.

The objective of PLM is to increase the product revenues, reduce product-related costs, maximize the value of the product portfolio, and maximize the value of current and future products for both customers and shareholders.

PLM eliminates waste and inefficiency across all aspects of a product's life, not solely in its manufacture. PLM is focused on using the power of information and computers to deliberately pare inefficiencies from the design, manufacture, support, and ultimate disposal of a product.

1.3 PHASES OF PLM:

Imagine	**Define**	**Realize**	**Use/Support**	**Retire/Dispose**

There are five phases in a product's lifecycle. In each of the five phases, the product is in a different state.

- During the imagination phase, the product is just an idea in people's head.
- During the definition phase, the ideas are being converted into a detailed description.
- By the end of the realization phase, the product exists in its final form (for example, as a car) in which it can be used by a customer.
- During the use/support phase, the product is with the customer who is using it.
- Eventually the product gets to a phase in which it's no longer useful. It's retired by the company and disposed of by the customer.

The product must be managed in all these phases to make sure that everything works well, and that the product makes good money for the company. That means managing the product throughout its lifecycle, **''from cradle to grave''**.
Three of the lifecycle phases (Imagination, Definition and Realization) make up the **Beginning-of-Life (BOL)** of the product. There is a **Middle-of-Life (MOL)** phase which includes activities such as product use, support and maintenance. There is an **End-of-Life (EOL)** phase which includes activities such as product retirement, disposal and recycling.

1.4 NEED FOR PLM:

Product development is the key to future innovation. But, the challenge here is that the new products take a long time to develop and reach the market. Without a system to manage all the different aspects of creating a new product, it is very easy for product development to run late and over budget.
PLM helps to make the product development process more transparent and improve efficiencies. This leads to more innovations, shorter product development cycles development and time-to-market and better determination of the status of new product development undertakings.
Thus in this context the need for PLM will be to

- Manage a well-structured and valuable Product Portfolio
- Maximize the financial return from the Product Portfolio
- Provide control and visibility over products throughout the lifecycle
- Manage products across the lifecycle
- Manage product development, support and disposal projects effectively
- Manage feedback about products from customers, products, field engineers and the market
- Enable collaborative work with design and supply chain partners and with customers
- Manage product-related processes so that they are coherent, joined-up, effective and lean
- Capture, securely manage, and maintain the integrity of product definition information and make it available where and when it's needed.
- Know the exact characteristics, both technical and financial, of a product throughout its lifecycle.

1.5 OPPORTUNITIES OF PLM:

PLM enables companies to take advantage of the many opportunities available at the beginning of the twenty-first century. Some of these opportunities are the result of new technologies. Others are due to social and environmental changes, and macroeconomic forces such as globalization.

Opportunities of globalization: Globalization has led many companies to operate in many countries. This has made it even more difficult to keep control of products. It has increased the number of potential customers falling under different ages for many companies. PLM will be needed to keep control of the products they acquire.
For example, weight reduction products, pharmaceutical products, cosmetic products, sports goods, drugs for domestic animals etc are available in market in multiple brands. As a global market competitor, the company should track its sales and do necessary modifications to sustain the market which is achieved by PLM.

Technology opportunities: New technologies open up new markets and lead to new products. Towards the end of the twentieth century, new technologies such as

nanotechnology, the control of matter at the Nano scale. They opened the way to even more new products. Down at the Nano level of individual molecules, matter can be arranged and assembled to make products such as new drugs, plastics and electronic circuits. Existing technologies such as electronics, computing, telecommunication, robotics and biotechnology continue to offer scope for new products. Designer drugs will be developed to match an individual's particular genetic make-up. Intelligent clothes will change performance as the weather changes and the wearer's mood changes. The Internet, the World Wide Web, Global Positioning System (GPS), Mobile telephony etc. offer opportunities for new products and services, and new ways to develop, sell and support products. Direct brain implants of memory and processing power will increase human performance. Sensors implanted in the body will monitor organ performance. RFID technology allows products to be tagged with chips that can provide information about the product when they are scanned. This allows products to be tracked throughout their lifetime. RFID offers opportunities to get a better understanding of the way products behave over their lifecycle. Environmental requirements and the desire for sustainable development will open up new opportunities, in particular for a product's end of life. Business models will change, with more and more services being offered around products. PLM will enable the development and support of new products and services.

Smart Product Opportunities: Smart Products, also known as Intelligent Products, are products that can sense and communicate information about their condition and environment. In addition to their primary functionality, they have functionality to decide or communicate about their situation or environment. They make use of sensors, synthesizers, motors, GPS, microprocessors, memory unit, and transmitter etc to satisfy their functionality. A Smart washing machine, equipped with a scanner, can read the labels on clothes, and select the most appropriate washing and drying cycle, packaging can be smart, Smart lawn mower can be programmed to cut the grass, its sensors see if there are any obstacles, identify the height of the grass, and switch on its motors to go down the garden and cut the grass, smart microwave oven can identify the food to be cooked, then set the timer and the temperature. A smart water softener can identify the hardness of incoming water, and treat it as required by its hardness and the intended use. As well as working independently, smart products can also work together. Voice applications include the scales that speak your weight and the voice in your car that reminds you to fasten your safety belt. Smart security devices can listen to your voice and look into your eyes before giving you access to a secure zone. Smart products have many applications and can work in dangerous areas to render them safe. PLM here can identify End-of-Life parts and materials that can be reprocessed or reused.

Social and environmental opportunities: The world faces increasing social problems. Society has always had an impact on manufacturing industry. Years ago, among the most highly visible effects of manufacturing industry were the factory chimneys and coal-burning fires that polluted cities, smog- a mixture of fog and smoke resulting from the combustion of coal that led to thousands in death. In addition, incomplete combustion of fuel leads to the production of carbon monoxide, a colorless, odorless, poisonous gas.. Initially, in the name of Progress, much is accepted, but eventually society catches up and legislates against dirty, poisonous products that kill and pollute. Over the years, population growth, lack of disposal sites, and scarce natural resources have led to all sorts of reduction, reutilization, recycling and recovery programs. PLM will play a key role in addressing all these issues because it provides the opportunity to get control of products across their lifecycles.

1.6BENEFITS OF PLM

a) Capture customer requirements better
b) Create more innovative ideas

c) Develop products in an International collaborative environment

d) Increase the number of customers by developing and supporting new products by reduced time to market and increase sales

e) Improved product quality and reliability

f) Reduced prototyping costs by reducing wastes

g) More accurate and timely request for quote generation

h) A framework for product optimization

i) Savings through the complete integration of engineering workflows

j) Documentation that can assist in proving compliance i.e. Savings through the re-use of original data

k) Ability to provide contract manufacturers with access to a centralized product record

l) Improved forecasting to reduce material costs

m) Maximize supply chain collaboration

1.7 DIFFERENT VIEWS FOR PLM:

Different stand point leads to different views i.e. different users see PLM in different views based on their implementation possibilities. The different stand points in the functionality of any industry includes CEO, top managers, product development managers, product support managers, engineering managers, quality managers, human resource managers, application system vendors etc. each of their views is discussed in brief below:

CEO: Implementing PLM is so important that the PLM initiative should be led by the CEO. However, CEOs have other things to think about than process mapping and the number of duplicate parts. The CEO has to set company directions, objectives, strategies, organizational structures, plans and budgets, lead the company, act as a figurehead for employees, shareholders and customers, and make sure that sales exceed costs. The CEO may have little time available for the PLM initiative, so whatever time that is available should be focused on activities where the CEO adds most value. While it's important to get the CEO to lead the PLM initiative, it shouldn't be expected that the CEO will get involved in the details. This could even be counterproductive, as CEOs often work with a very high-level abstraction of the business, with little understanding of the details of the product lifecycle. Sometimes they live in a slightly unreal world, not coming into contact with the people who develop, produce and operate their products. Thus PLM from CEO's view should be very much effective but can be extracted only after demonstrating the use of PLM by their competitor and their success rate.

Business planners: A PLM Initiative will cost a lot of money, and should bring a lot of benefits. With these characteristics, it will have to be included in company budgets and investment plans. Implementation of PLM needs better support from the people who look after the business plans. The business planners are meant to look after the figures and they

have ways of influencing decisions on proposed activities. They may have little hands-on experience of what happens throughout the product lifecycle, so it's unlikely that they will instinctively support PLM. It may well be that their only job in the company will have been a staff job. They may have little real experience of what happens in the product lifecycle, who the customers are, what the products are, or the processes that a product goes through from its conception to its use by a customer. Instead, they may have a spreadsheet view of the company – here are the costs, here are the sales, add some market share figures, some market growth, some tax rates.

Functional managers: Managers running a functional department, such as Engineering or Manufacturing, live in the real world, and are expected to produce instant results. As a result, they may not show much interest in benefits of PLM that may appear in successive years' time. They are often so busy working at day-to-day problems to meet the short term targets set by their CEO's that they cannot get involved in long-term issues. As a result, each may react differently to a proposal for PLM – each may only be looking for benefits in their particular function.

Marketing managers: Marketing managers will react positively to the CEO's interest in PLM. It will provide them a new reason to identify more finely segmented niche markets and imagine corresponding products. PLM can help Marketing Managers make sure better and faster responses are made to Requests for Proposal, material from successful proposals is re-used, and proposals are priced realistically and competitively. With all the product data and customer specification information available, it will be much easier for them to evaluate future opportunities. Manufacturing managers will claim they have been set the objectives of reducing costs and improving manufacturing productivity, and cannot see how these targets can be achieved by PLM. According to them, PLM should be the responsibility of the design engineers who create the products. In private, of course, the Manufacturing managers will agree that anything that gives some control over design engineers and product data must be a good thing. PLM can help Manufacturing operate more effectively by increasingly involved in product development activities, advising against difficult to manufacture designs, reducing scrap and rework.

Engineering managers and product support managers: Engineering managers know how difficult it is to develop new products in response to constantly changing requirements. They are used to the changes from Marketing and Manufacturing, and they try to improve their responsiveness. They know that sometimes they are not in complete control of the projects entrusted to them. Yet as they look round, they see the potential for improving the situation. They see time wasted as drawings are manually transported from one activity to another. They see time wasted as information waits to be signed-off. They see their engineers going into far too much detail where it is not needed, and not enough detail where it is needed. They see an unnecessarily high number of signatures holding progress back. They see marketing personnel wasting their time going far too deep into technical details. They see manufacturing personnel doing nothing as they wait for information. They see support engineers wasting time trying to use technical manuals that are out-of-date. They are aware of the scrap, the personnel time that is wasted, the possibilities of reducing the product development and support cycles. They know that late market entry is going to cost the company a lot of money. They understand as well as everyone else that the time to reduce product costs is during the development phase, when most of the product costs are defined. They understand that it is much cheaper to reuse, or slightly modify, an existing design than to start a new design from scratch. Like everyone else they know that it is important that engineering changes are

controlled and that exact product configurations are maintained. Most engineering managers will welcome PLM, unless their budget is going to be used to pay for it. Product support managers are at the end of the Marketing-Design-Manufacturing- Support chain in a company, and they face many problems generated earlier in the product lifecycle. For example, people in Field Service often have no way of knowing what's been configured in a particular product. Design engineers might send them an "as-designed" configuration, but all sorts of things can change after that is produced. Design Engineering may change components but not tell anyone. Some of the sub-assemblies may be purchased, and have different components to the initial in-house design. Manufacturing might also make changes. At the Assembly and Test stages, where there's always a lot of pressure to get the product out the door, additional changes may be made but not get properly documented. Most product support managers will welcome PLM, and will provide valuable input to the PLM Initiative.

Workers: Many of the workers in the lifecycle are highly-educated, well-trained, under-utilized, under-respected and under-paid. For years, they do their job well and see little reward, whereas other people rise rapidly up the company hierarchy. As functional organizations give people little chance to expand their skills horizontally by working in other departments, many workers eventually become isolated, highly focused on their own skills, and somewhat resentful of other departments. If they are not in Marketing, they see Marketing as over-indulged and over-paid in view of the poor product specifications it produces. If they are not in Engineering, they see engineers as uncommunicative theorists living in an ivory tower. If they are not in Manufacturing, they look down on manufacturing workers, often immigrants, who are incapable of making even the simplest product properly. Across the product lifecycle, the workers complain about bureaucracy and form-filling. They feel unable to use their talents and experience to the full. For example when product engineers had an idea they would talk to a colleague in Manufacturing, make a prototype, put it in their briefcase, and go off to see a potential customer. The customer would give some feedback, and soon there would be a new product bringing in sales of a few million dollars each year. But now – it's changed. Now, it's all paper and plans and gates and meetings and reviews and milestones and spreadsheets and job descriptions – and no-one can just go and see a customer with an idea for a new product. The product engineers hear a mixture of messages. Managers keep telling them to be closer to customers, and develop more products and services. They feel that would take more time and money, but they are told to reduce costs, improve quality and reduce lead times. At the same time, they are expected to fill in all the forms, and to meet the product development target dates in the company plan. As the individual requirements of the different requests are different, or even contradictory, product engineers have to walk a fine tight rope. They focus on what they think is most important, and most likely to be used by management to measure their performance, and look for ways to play down the other issues, or to pass them on to others. Probably there will be one or two key products that must be produced on time and to specification. Most of the effort will go on these. In such an environment, the people who work in the product lifecycle may not be very interested in whether or not PLM is introduced. They may understand the benefits it can bring, but may doubt that it will ever work as suggested.

Application system vendors: Native applications for product development create the product record which is stored in a centralized or distributed vault. The digital model, including options, in native or STEP-compliant format, can be viewed by team members from Purchasing and Marketing in a BOM-like structure from an advanced product configurator. Digital manufacturing is enabled by interoperability across multiple platforms and shrink-wrapped integrations to all leading state-of-the-art and legacy systems. Shop floor workers on

multiple sites can visualize tool paths prior to cutting metal. Service workers can navigate lightweight data models over wireless environments. The modular product architecture and data model offers customers all necessary functionality out-of-the-box tailored to the requirements of different industry sectors. System functionality includes collaborative product development; virtual engineering; system engineering; product data management; parts list management; configuration management; image management; document management; content management; engineering change management; bill of material management; complaint management; compliance management; standards management; process management; project management; program management; portfolio management; workflow management; requirements management; needs management; innovation management; sourcing management; knowledge management; enterprise content management; regulatory management; information management; menu management and recipe management. Some of the thoughts which generally arise about vendors are that they don't listen to company needs, but only try to sell their systems; they propose new technologies and standards for which they can't provide any business logic, some vendors claim basic functionality but can't even demonstrate this basic functionality. One vendor told him that they are looking for additional funding, and when they have completed the next round, they'll start developing the software.

1.8 COMPONENTS OF PLM:

Increasing numbers of manufacturers are utilising PLM solutions, to optimise all aspects of their product development processes. Nevertheless, it can be hard to offer a precise definition of PLM. Yet, to know exactly how PLM can benefit business, it is essential to know what PLM solutions have to offer. Regardless of how PLM is defined, there are the 'must have' components, which are vital for any PLM solution to be effective. We've put together an overview of the 7 essential components of PLM solutions.

 # 1 Document Management

This is vital for being able to store, track and manage all the data associated with your product
development processes. It includes everything from graphics and text right through to engineering calculations. Comprehensive document management makes it easier to track any data changes and monitor access to documents by creating a single, central data store.

 # 2 Embedded Visualisation

Embedded visualisation allows for collaboration, between mechanical and electrical areas, using centralised digital product information. It also makes it possible to view any product data, without needing authentication from the native tool.

 # 3 Workflow

This component of a PLM solution makes it possible to define product development processes fully, through a standardised method. It also ensures that all process and procedures are adhered to and all necessary related tasks taken.

 # 4 Distributed Collaboration

Companies increasingly operate across multiple locations, with internal and external partners. Distributed collaboration means both individuals and larger teams can work concurrently on a project, with all data securely protected, regardless of location.

 # 5 Multi-Cad Data Management

This enables complete control over all CAD data, from across different CAD tools. It offers a central point for all CAD data, while also managing the dependant CAD relationships during product development.

 # 6 Complete BOM Management

Manage and track all the different aspects of product development and definition. Complete BOM management covers the entire process, across all disciplines. It gives better control and offers engineers more insights into the potential impact of any changes.

 # 7 Change And Configuration Management

Good communication is essential, which is where change and configuration management comes in. It makes sure that everyone is aware of each process, from designing right through to servicing, throughout the entire product lifecycle.

1.9 FEASIBILITY STUDY:

The feasibility study is carried out to find out what type of approach and what level of response is appropriate for the challenges that the company faces.

● May be an enterprise-wide initiative targeting new market-leading products and control over the entire lifecycle is needed? If In which case it would be useful to develop a PLM Vision, a PLM Strategy and a PLM Plan.

● May be the main benefits can be achieved by implementing new lifecycle processes across several functions?

● May be there are benefits to be had by targeting some very precisely defined improvement areas?

The feasibility study should lead to a report of the form and contents shown below table

Title: Feasibility Study to Develop and Assess Options for our PLM Activity

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1.10 STRATEGIES

Strategy is the science of planning and directing large-scale operations, specifically (as distinguished from tactics) of maneuvering forces into the most advantageous position. Once

the PLM Vision has been defined, people will want to know what the organization will look like in five years. They'll ask how the resources in the product lifecycle should be deployed, structured and used in the next five years to achieve the Vision. They'll want to know how to organize the change from today's organization to the future organization.

Manufacturing strategy: For thousands of years, progress in increasing manufacturing productivity was slow, but a few hundred years ago, mechanization made a leap forward possible. Workers were assigned to a particular position at which they carried out a specific task. The owner supervised the workers making sure they worked at the pace of the machines. This led to a division of labour between the owner and the workers. The owner could not control all the workers all the time, so a hierarchy of supervisors and managers was developed. Machine tools changed the environment again, leading to strategies of mass production characterised by high volumes, mechanisation, organised material flow through various stages of manufacturing, sub-division of labour, and simplification and standardisation of common parts to allow long production runs of parts that could be fitted to other parts without lengthy adjustment. In mass production, tasks could be performed by unskilled workers, often immigrants or agricultural workers leaving the land, since much of the skill was in the machine and the organisation. Manufacturing enterprises grew to such a size that a large hierarchy of supervisors and managers became necessary. The increasing size and complexity of operations called for a large management staff of accountants, engineers, personnel managers, etc.

The next step, introduced to manufacturing by Henry Ford, was the assembly line. Each worker did one task, at a pace minimising unnecessary movement and increasing productivity.

Frederick Taylor brought a scientific approach to these principles. A new discipline – industrial engineering – appeared. Taylor broke each job down into its constituent parts, analysed them to find out which were essential, and timed the workers with a stopwatch. With superfluous motion eliminated, the worker, following a machine-like routine, became much more productive.

Mass production increased the trend to an international division of labour. Factories often needed raw materials from other countries, while saturation of national markets led to a search for customers overseas. The introduction of computers in Manufacturing in the mid-20th century led to strategies of shop floor automation (NC machines, CNC machines, robots, and flexible manufacturing systems). It also led to the introduction of MRP and ERP systems for planning and control of manufacturing and logistics. The logical ultimate in the evolution of strategies seems to be the re-configurable lights-out factory producing customised products in a batch size of one. This implies elimination of all manual labour and the introduction of flexible manufacturing and assembly machines with automatic controls providing accuracy and quality beyond human skill levels.

Company strategy: The strategies that companies adopt are also subject to change. Two main strategies have been used by companies to meet their objectives – a low-cost, "cost leadership" strategy and a strategy based on differentiation.

A cost-leader aims for the lowest product cost in a particular industry.This usually requires a high market share and a high volume of standard products, and implies substantial capital for large continuous-flow production runs and facilities.

Differentiation strategies are based on having a product or service that differs significantly (by virtue of its design, or technology, or customer service, etc.) from those of competitors.

Other strategies include "niche" and "trend-leadership". A niche strategy serves a particular market segment, or particular type of customer, or particular geography, or particular part of

a product range. Within a given niche, a company can hope to succeed with either a cost-leadership or a differentiation strategy. A company with a trend-leadership strategy will constantly innovate in an attempt to lead the market and be the first to produce a particular product or service, and gain the associated benefits. This type of leader is unlikely to be a cost-leader, due to the difficulty of getting products to market first.

Recent years have seen the introduction of new strategies such as "low-cost variety", "fast response time", "partnering", and process-based strategies (rather than product-based strategies) such as "capabilities-based competition" and Concurrent Engineering. The driving force behind many of these new strategies was Japanese companies using manufacturing excellence to gain competitive advantage.

Strategies such as "fast response time" have been introduced because, as a result of technological advances and changing customer behaviour, products have increasingly short lives.

"Partnering" is often driven by the growing need for innovation and the limited time available to develop new products. Partnerships between companies allow greater value and features to be offered to customers, while allowing each partner to concentrate efforts on things they do well.

PLM STRATEGIES

A strategy describes the way to achieve objectives; how resources will be organised, managed and used; policies governing use and management of resources. Deciding how to organise, manage and use the resources is a key part of strategy development.

A PLM strategy will be company-specific. Without knowing a particular organization in detail, it's not possible to say what its strategy should be.

A PLM strategy defines a way to achieve the Vision. It describes how resources will be organised and used. PLM strategies aren't generic. They are specific to individual organisations because they depend on the particular circumstances and resources of the individual organisation, and on its particular environment

Typically the PLM strategy will be developed for a five-year period. It will then be reviewed annually. The strategy should only be changed during the five-year period for very good reasons. Often it will take two or three years for the effects of a change of strategy to become evident

A PLM strategy is the starting point for developing and implementing improvement plans. A PLM strategy helps everybody to move forward along the same road towards the new environment.

Developing a PLM strategy is a five step process.

i. Collecting information

ii. Identifying possible strategies

iii. Selecting a strategy

iv. Communicating the selected strategy

v. Implementing the strategy

Collecting information: From the outline of the strategy document, it can be seen that a very good understanding of the activities and the resources in the product lifecycle is needed to develop the PLM strategy. This understanding must be based on factual information, not on guesses and opinions – which is why the first step of strategy development involves assembling the information with which the strategy will be developed. This information includes the vision and associated targets, customer and innovation objectives, details of

resources, capabilities and the environment, strengths and weaknesses, opportunities and threats.

Different techniques such as brainstorming, process analysis, and the use of questionnaires can be used to gather and understand the information describing the current situation. One of the ways to understand the current situation is to develop a questionnaire, and get the people who are supposed to be responsible for the individual topics to respond.
Another way of gathering information is to carry out an analysis of a typical product by following it through its lifecycle and asking questions where appropriate. This may give a less theoretical result than use of a questionnaire.
The use of product data, in particular at departmental boundaries, needs to be understood, as do data security and data integrity issues.
Information repositories need to be identified. They will contain bills of materials, manufacturing instructions, technical drawings, product specifications, CAD files, text documents and other forms of product information

Identifying the strategies: In the second step of strategy development, several potential strategies are identified, formulated and described in terms of the organisation and policies to be applied to the resources. It is always useful to identify and describe several possible strategies. This will improve the chances of finding the best strategy since the most obvious strategies are not necessarily the most appropriate. Some candidate strategies for PLM will be apparent from the organization's current and previous strategies, from those of competitors and other market players, and from reading about military and business strategies. In some cases, a candidate strategy will be closely linked to the company's business strategy. The strategy chosen for PLM has to meet the objectives of the company. As each company will have a different objective, as well as different resources and a different environment, the strategy a company develops will be different in some respect from that of any other company. It is to be expected that the strategy selected will contain some elements of one (or more) of the basic strategies, and also have some additional elements.
For example, a company may have a strategy of "customer focus". Because the PLM activity is an important activity, it will probably be expected to adopt this strategy. Company management may also set "customer focus" as an objective for the product development organisation. The product development organisation may respond to this objective with a strategy of "customer focus" or another strategy such as "fastest time to market". The product development organisation may also decide to adopt a strategy of "customer focus" to its internal customers within the company, for example to F&A and Service, and include their representatives in project teams.

Selecting a strategy: In the third step of strategy development, potential strategies are tested, and the most appropriate strategy is selected and detailed. It will be useful to investigate between three or four alternative strategies. This should lead to an in-depth understanding of the possible strategies. The strengths and weaknesses of a particular strategy often become clear when examining the strengths and weaknesses of other strategies.
The analysis of the different strategies is often called SWOT analysis. The acronym stands for strengths, weaknesses, opportunities and threats. Different authorities on strategy development use the SWOT acronym in different ways. Often the strengths and weaknesses apply to the organisation for which the strategy is being developed, and the opportunities and threats are those of the environment. The strengths and weaknesses can be those of the present or future situation, the opportunities and threats those of the current or future

environment. Strengths and weaknesses may also be defined for competitors. The SWOT analysis for PLM is carried out after identifying several potential strategies.
The questions are aimed at finding out which strategies are suitable, and which of the suitable strategies is the most appropriate. Among the questions to be asked are:

- Does this strategy meet the objectives?
- Is it in line with overall company objectives?
- Will it enable us to achieve the mission?
- How does it address key issues?
- How does it relate to key success factors?
- How will it be possible to measure progress towards specified targets?
- What would be the key metrics?
- How long would it take to implement?
- What are the overall costs?
- Do we have the financial strength to do this?
- What are the benefits to the organisation?
- Do we have the resources to do this?
- Can we find/afford missing resources?
- Does this strategy play to our strengths?
- Does it address our weaknesses?
- Does it allow us to take advantage of opportunities?
- Does it help us against threats?
- What will be the impact on customer responsiveness?
- Would this strategy save time?
- What would be the effect of this strategy on quality?
- What would be the effect of this strategy on product costs?
- What would be the effect of this strategy on the cost of product development?
- What would be the effect of this strategy on the cost of product support?

Communicating a strategy: In the fourth step of the five-step strategy development process, the chosen strategy is communicated to the people who will be affected or involved. Communication of the strategy is essential.
A strategy is useless unless the people who are going to be involved are fully aware of it, can understand it and can implement it. The name given to a strategy must be so short and simple that everyone (not just strategy theorists in ivory towers and company headquarters) can understand it.
A strategy has to be realistic. It has to be expressed in language that everyone can understand. There's no sense in including motherhood statements, or in claiming to greatly improve performance overnight.
It's important to have general agreement on vocabulary. Otherwise, if a group of people is asked to describe the strategy they will all give different answers. Before communicating the strategy, it should be made sure that everybody understands the meaning of the terminology.
Before disseminating the strategy, test it out on a few people who are typical of the audience. If they don't understand it, then they should be made understandable before communicating it more widely.
Decision should be made like to whom the strategy needs to be communicated to before starting to communicate it. Strategy should be communicated to everyone involved with PLM. The more they understand it, the more chance it has of being implemented. And also it should be made sure that the people understand the importance and necessary of strategy should see where it takes them as individuals.

Implementation of strategy: The PLM Strategy Report should contain a plan showing how the strategy will be implemented over the length of the total implementation (for example five years) and a more detailed plan for the first year. The structure for the plan should follow that for the PLM framework, making it easier for managers and others to see how all the issues are linked, and will be addressed.

- Start-up
- Management of the initiative – Project management
- Organisational structure of activities across the extended enterprise
- Processes
- Organisational structure – internal
- People
- Information
- Working methods
- Information systems
- Accompanying actions

Different views of the plan should be prepared – with different levels of detail. The first view should be a block diagram only showing which types of activity will take place in each year as shown in below table. This level of detail may be sufficient for company management. Other views of the plan will show more details of the activities. They will be needed for people who participate in, and manage, the activities.

	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5
Start-up					
Management of the initiative – Project management					
Organisational structure of activities across the extended enterprise					
Processes					
Organisational structure – internal					
People					
Information					
Working methods					
Information systems					
Accompanying actions					

STRATEGY ELEMENTS:

The exact meaning of a strategy element will differ from one company to another. For example, the strategy elements of "fastest time-to-market" and "lowest-cost competitor" could both be implemented in many ways. "Fastest time-to-market" could be implemented by building up a pre-defined stock of solutions, by increasing the number of engineers, or by shortening the product development process by removing non-value-adding activities.
Strategy elements can be described in exhaustive detail. Some of the key points could be:

Customer involvement

- Customer involved in product development and support teams
- Frequent communication with the customer

- In-depth customer surveys
- Specifications negotiated and optimised with customers
- Development and support work on customer sites (even in other countries)
- RFID integrated into product
- Online feedback from customer use.

Customisation capability

- Perceived by customers as excellent at customisation
- Able to identify customisation requirements quickly
- Able to identify customisation requirements correctly
- Able to carry out requested customisation quickly.

The highest functionality products and/or service

- Perceived by customers as the market leader
- Market expectation of continued functionality leadership
- Unique product and/or service
- Technological leadership
- Premium features.

The most robust product or services

- Product has long mean time between failure
- Product requires little maintenance
- Extensive simulation and testing
- Simple product or service with small number of parts
- Product certification demonstrating robustness.

The best processes

- Focus on implementing the best development process and methodology
- Able to do tasks well
- Customer belief in process competence
- Process certification.

Fastest time to market

- Projects started early with up-front effort profile
- Information shared as early as possible
- Simulation rather than test
- Information re-used.

Value-adding lifecycle

- Move up the added-value chain
- Focus on key value-adding activities
- Excellent reputation for key activities
- Activities that add little value are outsourced.

Lowest-cost competitor

- Simple product or service

- Standardisation
- Reuse

The longest life product

- Modular design
- Easy to upgrade
- Add-on services

Environment-friendly products

- Sustainable development
- No pollution
- Reuse.

1.11 PRODUCT DATA MANAGEMENT (PDM) SYSTEMS:

Product Data Management (PDM) systems are one of the most important components of a PLM solution. They are the primary system component of PLM. They are the systems to manage product data and product workflow.

The basic components of a PDM system include:

- The information warehouse or vault. This is where product data is stored.
- The information management module, which manages the information warehouse. It is responsible for such issues as data access, storage and recall, information security and integrity, concurrent use of data, and archival and recovery. It provides traceability of all actions taken on product data.
- The user interface. This provides a standard, but tailorable, interface for users. It supports user queries, menu-driven and forms-driven input, and report generation.
- System interfaces for programs such as CAD and ERP.
- Information and workflow structure definition functions which are used to define the structure of the data and workflows to be managed by the PDM system. The workflow is made up of a set of tasks. Data such as resources, events, responsibilities, procedures and standards can be associated to these tasks.
- Information structure management functions that maintain the exact structure of all information in the system across the product lifecycle
- Workflow management functions that keep workflow under control, for example, managing engineering changes and revisions
- System administration functionality which is used to set up, and maintain, the configuration of the system, and to assign and modify access rights.

PDM systems are used in the management of the activities in the product lifecycle. These activities start with the specification of a product and include product definition, analysis, manufacturing engineering, shop floor activities and product support.

PDM systems offer the potential for better use of resources, better access to information, better reuse of design information (since this will be under better control), better control of engineering changes, a reduction in development cost (since it will be easier to be aware of real costs during the engineering phase), a reduction in lead times, and improved security of product information.

PDM systems help companies to improve their competitive edge. They help improve the productivity of the product development process. They allow companies to be more flexible in their manufacturing. They help companies improve the quality of their products, and they

allow these companies to be more adaptable to market requirements, and more supportive of customers.

PDM will improve product development productivity. Product development managers will know the exact design status. They will be able to assign resources better, and release designs faster and with more confidence. Design engineers will know which parts are available and which procedures should be followed when designing new parts. Manufacturing engineers will be able to see how similar parts have been made previously PDM systems improve the way that large numbers of people, co-located or distributed, can work together. As a result, they have a visible effect on parameters such as lead time.

1.12 IMPLEMENTING A PDM SYSTEM:

Anyone implementing a PDM system is likely to be faced by people who don't want to change the status quo. There are lots of people in most companies who can provide all sorts of justifications for not doing whatever is proposed. To prepare for such behaviour, anyone who wants to implement PDM needs to be well-armed with reasons. Fortunately, there are plenty of them.

The reasons for implementing a PDM system can be divided into two classes. In one of these, the PDM system appears to alleviate some of the problems that occur in the product development environment. In the other, it appears proactively and positively impact operations across the product lifecycle.

The first class of reasons includes those in which the PDM system is used to overcome currently existing problems. Many of these reasons are related to the rapidly increasing amount of data in the product development environment. There is a general trend towards product development operations becoming more information intensive, and a corresponding push to use information-driven processes to support overall business activities. The increased computerization of product development activities leads to an increase in the volume and availability of product data. Without PDM systems this information can't be managed effectively.

Among the reasons for PDM that fall in the category of positively impacting operations are: better use of resources, better access to information, better reuse of design information, better control of engineering changes, a reduction in design cost, a reduction in lead times, and improved security of product information.

The two classes of reasons for using PDM systems are very closely linked. Reasons in the first class address the resolution of existing problems. Reasons in the second class go one step further, and address the potential for further improvement. The reasons can be grouped into eleven categories. In each category, most of the reasons can be related both to the resolution of current problems and to proactive improvement of activities across the product lifecycle.

Information management

- Provide a single, controlled vault for product information
- Maintain different views of information structure
- Provide faster access to data
- Manage configurations

Reuse of information

- Make available existing designs for use in new products
- Reduce duplicate data entry

Workflow management

- Make sure the most appropriate design process is followed
- Improve distribution of work to engineers
- Ensure release procedures are followed

Engineering change management
- Speed up the distribution, review and approval of engineering changes
- Provide status information on engineering changes

Overall business performance improvement

- Improve product quality
- Reduce overhead costs

Resolution of business problems
- Reduce scrap
- Reduce product liability costs

Functional performance improvement
- Increase engineering productivity
- Reduce inventory
- Develop better cost estimates

Better management of product development activities
- Improve project co-ordination
- Increase the reliability of product development schedules
- Provide high-quality management information

Automation of product development activities

- Automate the sign-off process
- Automate the transfer of data between applications

Information systems effectiveness improvement

- Integrate Islands of Automation
- Link databases together
- Remove unnecessary systems

Infrastructure for effective product development

- Support product development practices and computer systems
- Distribute data, documents and messages electronically.

The many very good reasons for doing PDM should not be allowed to mask the difficulty of actually achieving real benefits from its use. Benefits will not occur just because a PDM system has been bought and installed. Many other actions will be needed, such as setting clear targets for system use, deciding how to manage the system, training people to use the system, training the system support team, defining system use procedures, clarifying information structures, agreeing on information item ownership, defining the best overall product development process, modifying workflows, and defining and using a suitable product development and support methodology.

1.13 Differences between PLM and PDM

• PLM manages product information within a database

• PLM specifies and controls the complete, approved engineering design: requirements, specifications, procedures, configurations. It defines the current product structures and planned changes, as well as maintains the history of all previous design decisions.

• PLM data is usually created and managed by engineering in cooperation with other product managers in purchasing, production, quality, service and sales. It's the container for how to buy, fabricate, assemble, test, calibrate, inspect, install, repair and even sell the end product. PLM is a cross-department information storehouse, and its data is often exported to manufacturing systems and supply chain partners.

• PLM manages objects – parts, documents, change forms, and supporting data – within a database. These objects have:

• Descriptive attributes like owning organization, identifying number, name/title, revision (technical content) and lifecycle (business rules for what can be done with that content), weight and unit of measure. CAD files (and any other files) can be attached to database records to further describe the object.
• Relationships to other objects: parts have requirements, specifications, inspection procedures, etc.; assemblies have components with quantities; purchased parts have approved sources; designed parts have design drawings.
These attributes and relationships are created, reviewed and approved using system rules and change workflows.

Question bank:

1. Define PLM and explain the need for PLM.
2. Define PLM. List and explain different phases of Product Life Cycle.
3. Define PLM and List out the benefits of PLM.
4. List and explain the Phases of PLM.
5. List and explain the components of PLM.
6. Describe the PLM Feasibility study.
7. What is PLM strategy? Explain.
8. Sketch and explain the PLM has a holistic approach to the management of a product.
9. List and explain the various opportunities of PLM.
10. What do you mean by PLM feasibility study? Explain.
11. Define PLM strategy? Explain selection and implementation of Strategy.
12. Explain Product Data Management system and its implementation.
13. Describe the term Product Data Management and list out the advantages of PDM.

MODULE 2: PRODUCT DESIGN

This phase includes the specifically design-related activities, from the definition of product requirements and concept generation to the translation of the latter into a producible system.
Product development process the specific design phase is divided into a further sub process (the product design process), which transforms the set of functional specifications and product requirements into the detailed description of the constructional system interpreting them.
This transformation is achieved through a design route that begins with the definition of the problem and identification of the requirements, continues with the definition of the product concept, and concludes with the detailed specification of a producible design solution.
Product design is understood as a process whereby an organizational structure defines a problem and translates it into a feasible solution, making a series of design choices that each depend on the preceding choices and on a set of variables that collectively define the product, how it is made, and how it functions.

2.1 Engineering Design

The **engineering design process** is a common series of steps that engineers use in creating functional products and processes. The process is highly iterative - parts of the process often need to be repeated many times before another can be entered – though the part(s) that get iterated and the number of such cycles in any given project may vary.
It is a decision making process (often iterative) in which the basic sciences, mathematics, and engineering sciences are applied to convert resources optimally to meet a stated objective. Among the fundamental elements of the design process are the establishment of objectives and criteria, synthesis, analysis, construction, testing and evaluation.
In engineering science, design is the activity that enables the creation of new products, processes, systems, and organizational structures through which engineering contributes to society, satisfying its needs.
As with the product development process, the engineering design process cannot easily be assigned a single common scheme due to the great variety of possible design experiences. To summarize this variety, some authors distinguish between product design processes according to the principal categories of design intervention.

Creative design—This typology includes design studies constrained by specific requirements (functionality, performance, producibility) but with no specifications regarding the transformation of the idea into product or the realm of possible solutions.

Innovative design—In this case, the overall design problem and its possible decomposition into simpler sub problems is already known. Intervention then consists of synthesizing the possible alternatives for each constructional subunit, and can be reduced to a simple originatively combination of pre-existing components.

Redesign— This category includes interventions altering and improving pre-existing designs. This is necessary when a product does not fully meet the prescribed requirements or when changes in the environmental context for which the product was destined produce new requirements to which the product must be adapted if it is to remain on the market.

Routine design—In this case, different characteristic design factors such as the form of the product, the method of design approach, and the production system are all known before the

design process begins. Intervention is then reduced to the choice of the best alternative with respect to each subunit of the product.

PRODUCT DEVELOPMENT PROCESS

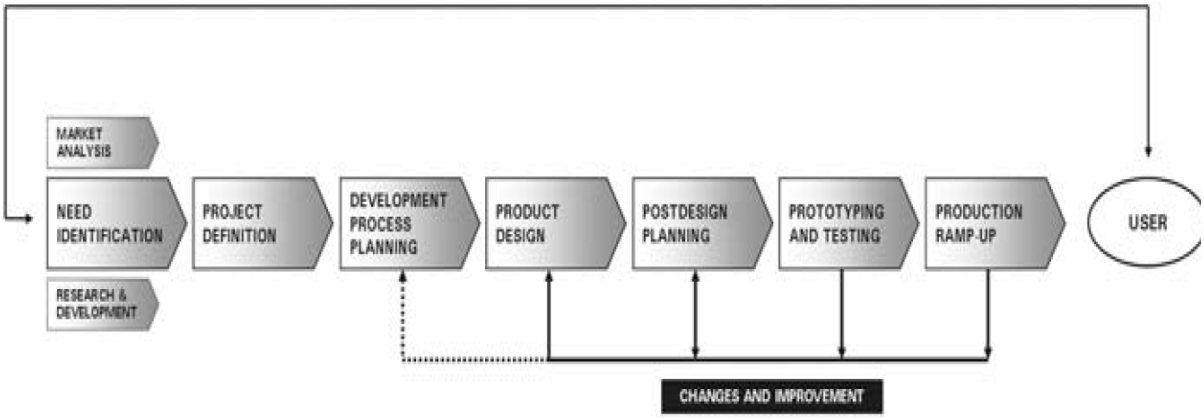


Fig 2.1: Product development process

Need identification—This phase consists of acquiring information on the needs of the customer, identifying the user typology and competing products on the market, and evaluating the most appropriate strategy (improvement of a pre-existing product, development of new technologies). Precisely to highlight the close ties this phase has with knowledge of the market and the opportunities afforded by technological innovation, the parallel activities of market analysis and research and development are also included in the above fig.

Project definition—This is the phase where the project is approved, and constitutes the true and proper beginning of product development. It summarizes the company strategies, market reality, and the technological developments in what is called the project mission statement, which describes the market goal of the product, the company objectives, and the main restraints of the project.

Development process planning—This phase involves planning the entire design and development process, through the decomposition, planning, and distribution of activities, the definition and distribution of resources (temporal, financial, and human), and the acquisition and distribution of information.

Product design—This phase includes the specifically design-related activities, from the definition of product requirements and concept generation to the translation of the latter into a producible system. This phase is, therefore, in turn divided into a further sub process, the design process, which will be discussed in the next section.

Post design planning—This generic title is used to indicate the specific phase regarding the planning of the production–consumption cycle.

Prototyping and testing—This phase requires the development of product prototypes that are then tested in order to evaluate how well the proposed solution satisfies the prescribed requisites, the performance levels offered by this solution, and its reliability. Clearly, this

phase has a preeminent influence on interventions to improve the product and, therefore, on the evolution of the design solution.

Production ramp-up—This phase of starting up production completes the product development process. It consists of manufacturing the product using the planned production system (this is not the case in the prototyping phase, where the product is created in another way). The main aim is to verify the suitability of the real production process, to resolve any problems that arise, and to identify any remaining defects in the final product. Ramp-up is then followed by a phase of transition toward actual production and the definitive launch of the product on the market.

ORGANIZATION AND DECOMPOSITION IN PRODUCT DESIGN

The phase of development process planning consists of the decomposition, planning, and distribution of all the activities, resources, and information involved in the entire process under consideration. This phase plays a determinant role in relation to the ever-increasing complexity of design problems and sees the emergence of the viewpoint known as the operations management perspective, where the product is perceived as the set of activities necessary for its manufacture and marketing. This viewpoint is implemented through the application of the general principles of Organization Theory, wherein the particular function of product development consists of the planning and organization of design and development activities (i.e., organization of the design and development process).

Underpinning this specific function of design process organization (which by its very nature encompasses both engineering and managerial competences) is the now commonly used practice of modeling the product design process—breaking it down into single tasks and determining the structure and the interactions linking these together. This decomposition makes it possible to reduce the design problem to simpler sub problems, and thus becomes an approach to the management of the ever-greater complexity of design

Furthermore, the study of individual design tasks can be an effective approach to the analysis of alternative design strategies, and ultimately to an improvement of the overall design process

The importance of organizing and managing the design process is thus clear. This process must then be supported by three different typologies of knowledge: knowledge to generate the ideas, to assess the ideas, and to structure the design process itself.

Within the study of the decomposition of the design process, there is a distinction between the two disciplines most involved in product development—engineering design and management science. In engineering design, attention is directed at the structure of the constructional system (study of the product architecture); management science focuses on the structure of the organization managing the project (study of the division and organization of the activities).

In a complete perspective, the principle of decomposition can be extended to three different domains: product, process, and organization. In the product domain, decomposition consists of splitting the complex system into subsystems, subassemblies, and components. In the process domain, it consists of dividing the design process into tasks, activities, and work units. Finally, in the organization domain the decomposition involves structuring the human resources into teams and workgroups and assigning them individual tasks.

PRODUCT DESIGN PROCESS

Design becomes the instrument linking functional requirements (which are part of the functional domain) to the physical solution (characterized by design parameters and belonging to the physical domain). The process of product design begins, therefore, with

defining the functional requirements that satisfy a given set of needs and translating them into design parameters. Product design finishes with the creation of the physical object satisfying these requirements.

"Product design" is understood as an activity that applies scientific techniques and principles to a set of information (needs, requirements, constraints), with the aim of defining the constructional system comprising the product in a manner sufficiently detailed to allow its physical realization.

The product design process then becomes a process of transforming information from the state where it describes and characterizes a product demand (needs, constraints, consumer requirements, market conditions, available technology) to a state where it fully describes the technical systems able to satisfy the initial demand. This process of transformation is achieved through the use of various types of resources (cognitive, human, economic, material) that fuel the main phases of the design process.

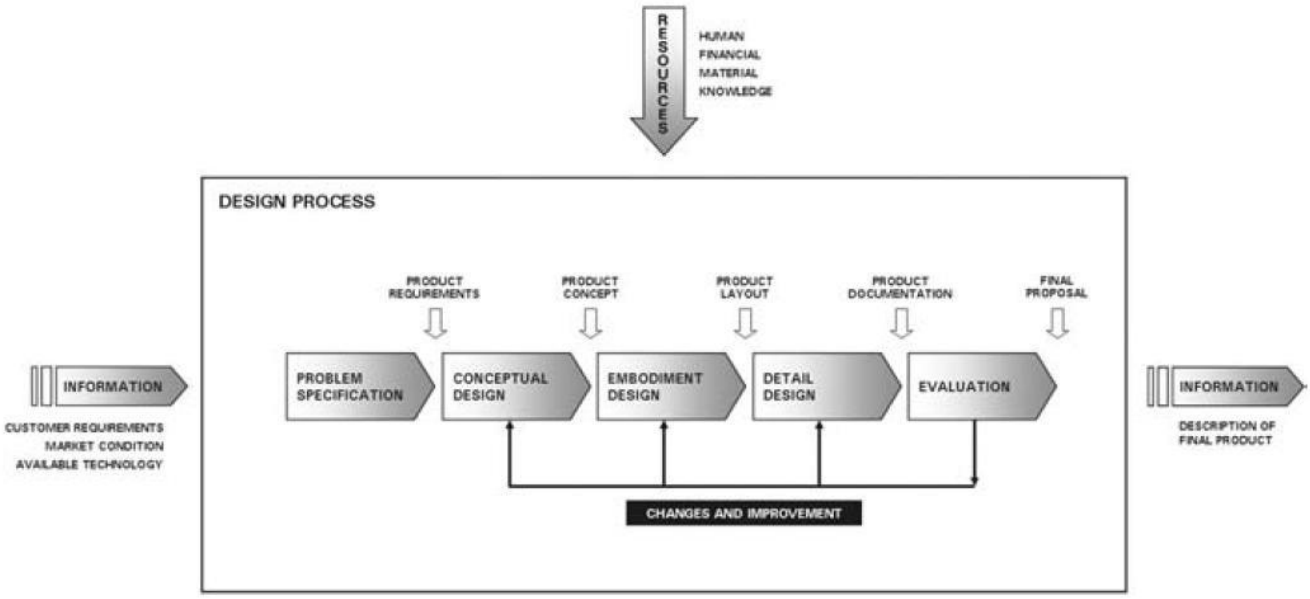


Fig 2.2: Product design process

The different phases of product design process are a preliminary phase of specifying the problem and defining the product requirements and phases of design at different level (concept, system, detail), of assessment, making up the iterative cycles of analysis–synthesis–evaluation.

Problem Specification—In this phase all the information relative to the project in question is elaborated to develop and define the requisites that must characterize the product. Information describing the needs to be satisfied, the consumer requirements, the market conditions, and the company strategies must be clarified (and integrated, if necessary) and used to generate the specifications that will guide the subsequent design phases.

Conceptual Design—Having defined the project specifications, it is necessary to develop ideas that will allow the creation of a product with the desired requisites. In this phase, "product" is understood in the abstract sense, as a set of attributes that must be embodied in the product concept. This is achieved through a first step of generating ideas (concept generation) and a second step of assessment and selection (concept evaluation) (Dieter, 2000; Ullman, 2003). In fact, a certain number of general concepts are proposed (description of the

shape, functions, and main characteristics of the product) and then assessed in order to determine which of them best meets the intended target.

Embodiment Design—Having identified the most appropriate concept, the next phase is the preliminary interpretation of the design idea in a physical system. The concepts formulated in the previous phases are developed, their feasibility is verified, and finally they are translated into a general product layout that defines subsystems and functional components. In this phase, the physical elements are combined to achieve the required functionality and the product architecture thus takes shape. This phase also includes a preliminary study of the shape of the components and a first selection of materials.

Detail Design—The layout developed in the previous phases must be translated into geometric models and detailed designs. This requires the application of methods and tools aiding a correct definition of the design details. The choice of materials, study of the shapes, definition of the geometry of components and assemblies, and the development of the assembly sequences and definition of the junction systems must all be guided by the entire range of product requirements (performance, economic, environmental, etc.)

METHODOLOGICAL EVOLUTION IN PRODUCT DESIGN

In general, the apparent complexity of the design process and of the system of factors influencing it in various ways is counterbalanced by a limited freedom of choice over the materials to be used and the ways in which these materials can be processed and assembled in the product in order to obtain the required functions. This condition complicates the intervention of the designer, whose objective is to balance the various specifications of function, cost, and reliability to achieve the appropriate compromise allowing the attainment of an ever-broader spectrum of required performances. As a direct consequence, the design process and production management have required considerable innovation in recent years, principally with the aim of reducing the time and resources employed in the design, production, and distribution of products. In this context, one of the more important considerations is the need to extend the vision of the problems to be taken into account to cover the entire product design and development process. This involves moving from a conventional approach limited to considering the sale of the product as the final step of the analysis, to an innovative approach where the phase of product use is also considered, going so far as to include the product end-of-life and disposal.

The product design and development models described above in Fig 2.1, initially characterized by rigidly sequential structures, must therefore be inserted in new methodological contexts that provide for design actions of analysis and synthesis that are simultaneous and in close interaction, in relation to all the phases of product development.

The sequential model, represented schematically in Figure 2.1, thus evolves into the simultaneous/integrated product development model as shown in Figure 2.3.

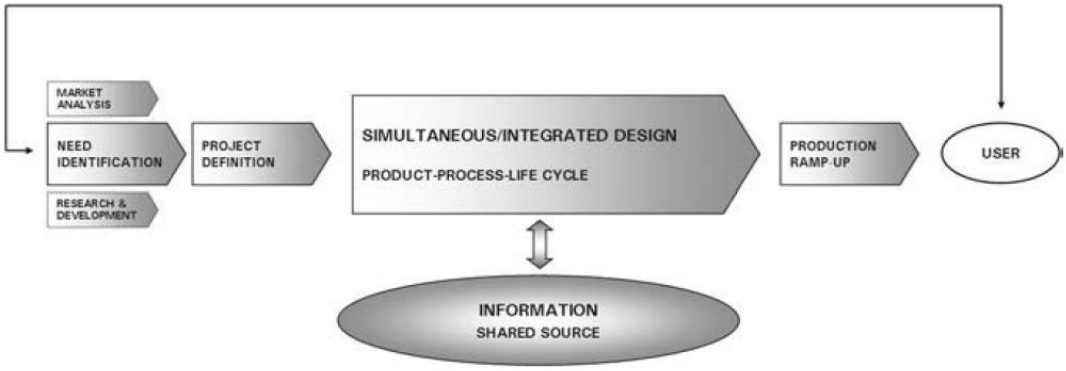


Fig 2.3: Simultaneous/integrated product development process

In this model, the phases of process development planning, product design, production–consumption cycle planning, and results evaluation are fused in a single, simultaneous intervention that draws information from a shared source and takes into account a wide variety of aspects (functionality, producibility, reliability, and cost).

In the context of this new, simultaneous/integrated statement, three approaches are currently the subject of much of the research regarding design methodologies:

• Concurrent Engineering (CE)—Aims at a full harmonization between the increase in product quality and the reduction of development times and costs through a structuring of product development that involves a large design team conducting simultaneous and interconnected analysis and synthesis actions, in relation to all the phases of development.

• Design for X (DFX)—Involves a flexible system of design methodologies and tools, each directed at the attainment of a particular product requirement.

• Life Cycle Design (LCD)—Extends the field of design analysis to the entire life cycle of the product, from the production and use of materials to disposal.

CONCURRENT ENGINEERING

It can be defined as a systematic approach to the integrated and simultaneous design of products and processes that includes production problems and user support.

The aim of this approach is to have, from the very beginning; the product development team simultaneously considers all the determinant factors operating on the product's life cycle, from concept development to retirement, including the requisites of quality, cost, and production planning, together with the exigencies and requests of the user.

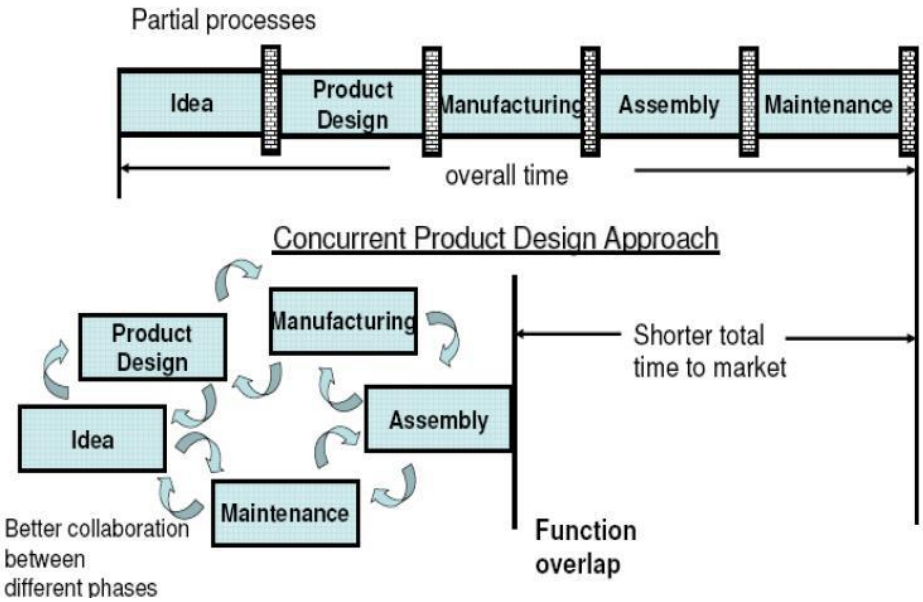
Reducing product development times is one of the primary objectives of CE. It is easy to imagine that one positive effect resulting from the simultaneous approach to design intervention is precisely that of shorter times.

The below figure compare the structure of the traditional process of product development with that of a simultaneous or concurrent development process. It should be noted, however, that the integrated approach leads to an increase in the complexity of the design problem, which may itself prolong development times. The primary objective of CE is, therefore, to reconcile this increased complexity, due to the level of integration between design and other company functions (principally that of manufacturing) with the control and reduction of the times and costs of development and production.

Furthermore, the possibility of overlapping the activities of the design and development process, in order to make it as simultaneous and integrated as possible, depends on how each of these activities is correlated with the others. Some activities may be completely independent of the others and can thus be performed simultaneously. Others will require information produced by activities that clearly must be performed first, thereby setting up a

sequential structure. Other activities can be intimately linked, with one requiring the information produced by the other and vice versa, so that they must be performed iteratively. Thus, the possibility of rendering the design process simultaneously is strictly dependent on the decomposition and planning of the design activities.

The continually broadening spectrum of design requirements and the increasing importance of producing customer-oriented products, together with new demands associated with the phases of product retirement and disposal is achieved through the life cycle approach, which promotes the consideration of not only the necessities of production but also those of the subsequent phases of the product's life cycle. The life cycle approach significantly broadens the range of factors to be taken into account: functionality, manufacturing, assembly, testing, maintenance, reliability, cost, and quality.



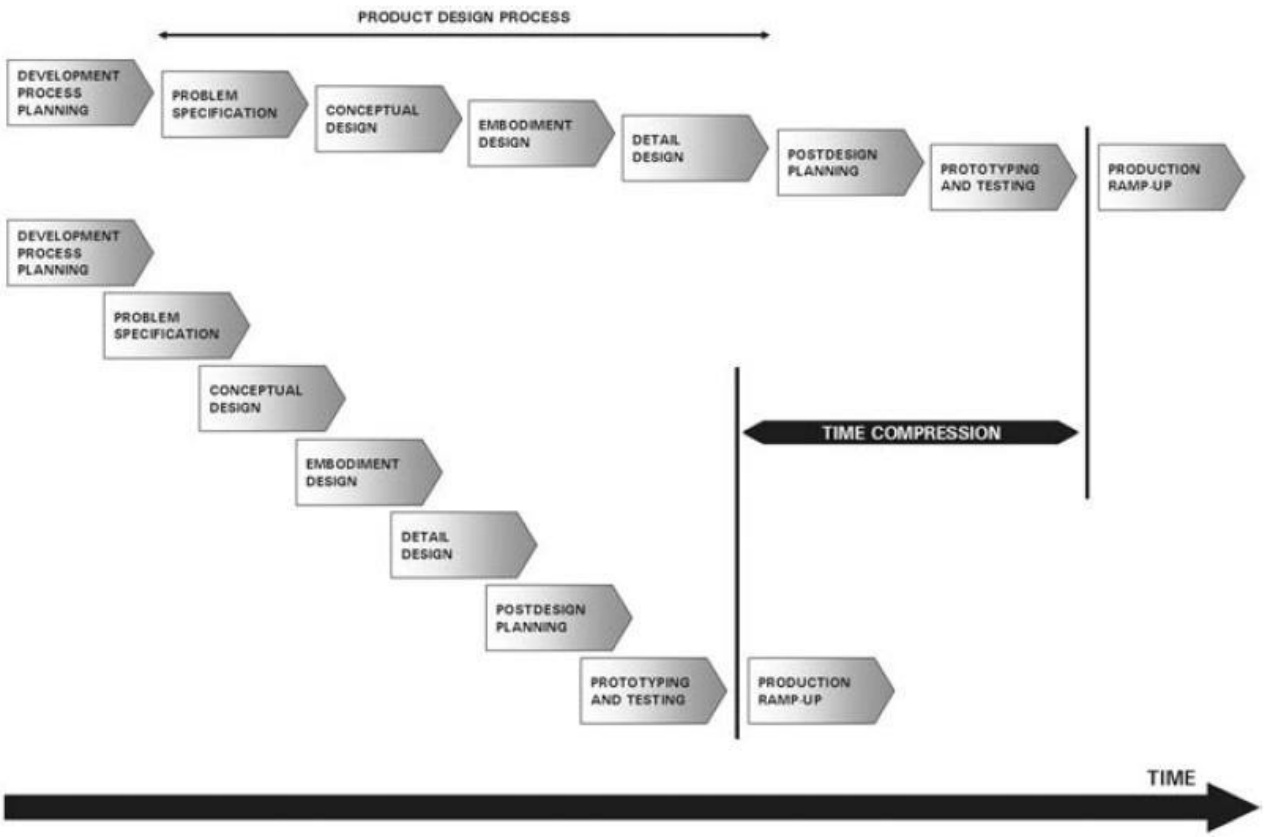


Fig 2.4 Comparison between sequential engineering and concurrent engineering

This aspect clearly brings Concurrent Engineering closer to Life Cycle Design (LCD). LCD is a design intervention that considers all the phases of a product's life cycle (development, production, distribution, use, maintenance, disposal and recovery).

DESIGN FOR X (DFX) AND DESIGN CENTERED DEVELOPMENT MODEL

Here X stands for different properties of the product, characterizing it in relation to one or more phases of its life cycle. Generally, DFX is an approach to product design expressly directed at maximizing the requisites demanded of the product (functionality and performance, manufacturability, quality, reliability, serviceability, safety, user friendliness, environmental friendliness, short time to market) and, at the same time, at minimizing costs.

Since these requisites characterize the behavior of the product over its entire life cycle, these premises have led to the identification and classification of criteria for design oriented toward the production, assembly, use, and recovery of the product, with the objective of developing an aid to design as complete and flexible as possible that takes into account the exigencies deriving from all the phases of the product’s life cycle.

Ultimately, DFX becomes a design system made up of tools and techniques aiding decision making, conducting different types of analysis and, through appropriate metrics, quantifying the performance of the design choices.

These tools can assume different forms, from a set of guidelines to a detailed procedure using analytical models, sometimes implemented in software program.

Some of the main objective properties (and, therefore, also the corresponding tools and techniques of the DFX system) can be grouped in relation to the phase of the product’s life cycle where they are most precisely applied.

Regarding the production phase, the most common DFX techniques are:
Design for Producibility/Manufacturability—Design of components and constructional systems as a function of machining processes.

Design for Assembly—Design for facilitating assembly, reducing assembly times and the possibility of error.

Design for Variety—Design oriented toward product variety, achieved through definition and optimization of basic model typologies, to provide a flexible product architecture guaranteeing variety in relation to market demand

Design for Robustness/Quality—Design of components directed at reducing the sensitivity of product performance to uncontrollable disturbance phenomena (machining errors and unforeseen operating conditions), and guaranteeing product quality

In relation to the use phase, there are:
Design for Reliability—Design of components and systems aimed at improving the reliability of the constructional unit

Design for Serviceability/Maintainability—Design oriented toward facilitating servicing and maintenance, to favor repair or interventions directed at excluding the possibility of failure phenomena (e.g., the replacement of worn parts)

Design for Safety—Design oriented toward controlling safety standards and the prevention of malfunctions during use

In relation to the end-of-life phase:
Design for Product Retirement/Recovery—Design directed at the planning of disposal and recovery strategies at the end of the product's useful life.

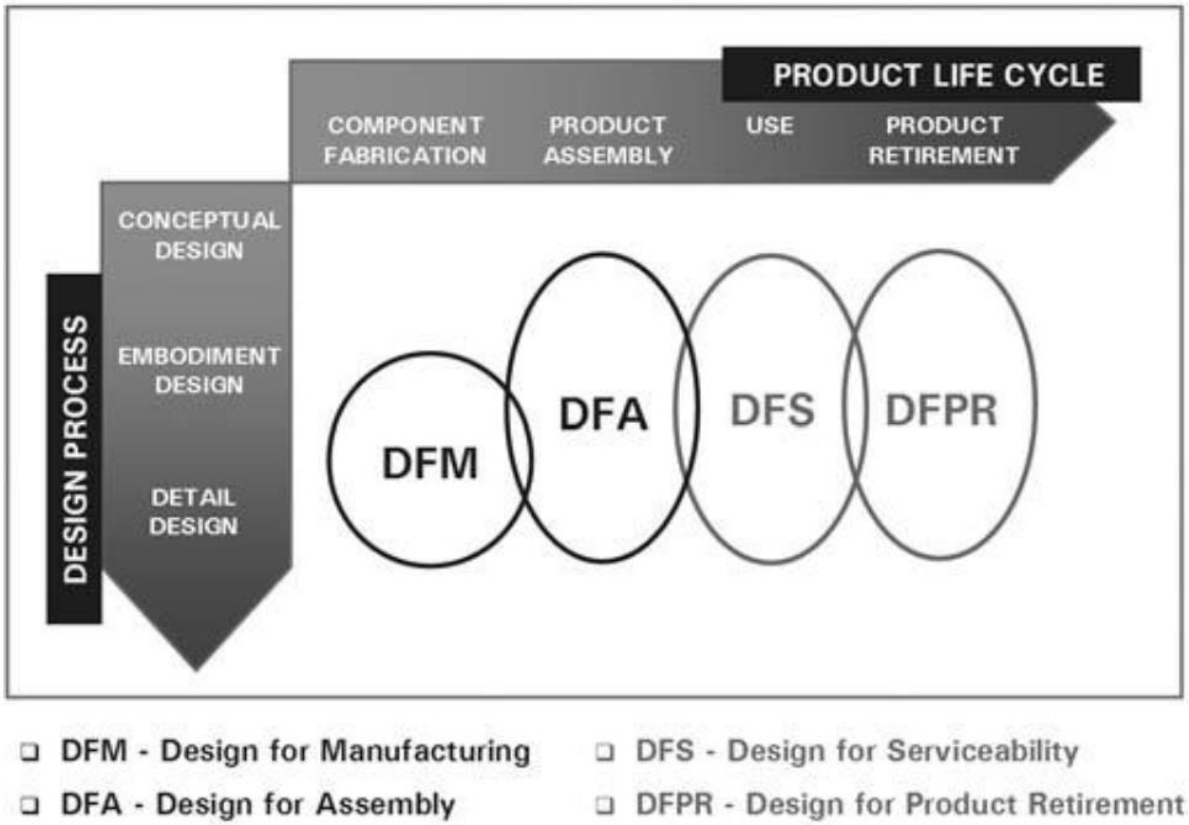


Fig 2.5 an example for the use of Design for X

DFX is a system of tools and techniques for design which, if suitably structured and managed, can be used effectively in the product development process, permitting the improvement of a wide range of aspects in industrial production, including product quality, costs, and time to market.

Although DFX can thus be considered an approach allowing the implementation of CE (focusing on a limited number of aspects at a time), in the context of CE its tools are not considered essential because their contribution is provided by the direct competence of the multidisciplinary.

The essential difference between the design-centered model (to which the DFX system is grafted) and the concurrent model at the root of CE lies precisely in the fact that they are based on different strengths: in the first case, it is the greater effectiveness of the specifically designed-related phases and of decision making; in the second, it is the greater competence found in the development team.

STRATEGIES FOR RECOVERY AT THE END OF LIFE, RECYCLING IN PRODUCT DESIGN

Products destined for reuse, in their entirety or only in part, can be collected and directed at the same original use or, alternatively, at another use demanding lower performance standards and having less-stringent requisites.

The operations required may be limited to the cleaning or disassembly of some parts and their reassembly in new products. With regard to the possibility of remanufacturing (i.e., the process of reconditioning products worn through use and returning them to their almost original condition), it is especially important to facilitate the removal and substitution of parts and components and favor their interchangeability within the same line of products.

When the objective of disassembly is recycling, the condition of equilibrium defining the profit margin and economic sustainability is sensible to variations in the price of virgin materials and in the costs of collection and recycling.
A complete panorama of the entire range of possible recovery strategies can be structured by clearly distinguishing between strategies for the recovery of the entire product and those for its parts or components.
Starting from this presupposition, at the end of a product's useful life the recovery, intervention can follow one of the following different routes

- If the product is in optimum working condition and guarantees the functional standards, it can be directly reused (in the figure, Reuse of Product).
- If it is in good condition it can be reprocessed, reconditioned, and reused (Remanufacturing of Product)
- If it is entirely composed of recyclable and compatible materials, it can be fully recycled (Recycling of Product).
- If the product contains parts that can be reused, reprocessed, or recycled, it must be disassembled. This operation can, in turn, lead to four different possibilities of recovery: reusable parts can be detached from the product, returned to the production process, tested and reassembled in new products (Reuse of Components); other parts can be reprocessed before being reassembled in new products (Remanufacturing of Components); parts unsuitable for the two previous solutions and composed of recyclable materials can be directly routed to material recycling (Recycling of Components) parts containing volumes of recyclable materials can be dismantled and the resulting materials separated into volumes for recycling and volumes for dumping (Dismantling, Shredding, and Recycling of Component)
- If the product contains small volumes of recyclable materials, it can be dismantled to allow the subsequent selection and recycling of these materials (Dismantling, Shredding, and Recycling of Product).
- Finally, if the product does not contain reusable or recyclable parts it is directly routed to the waste disposal site (Disposal).

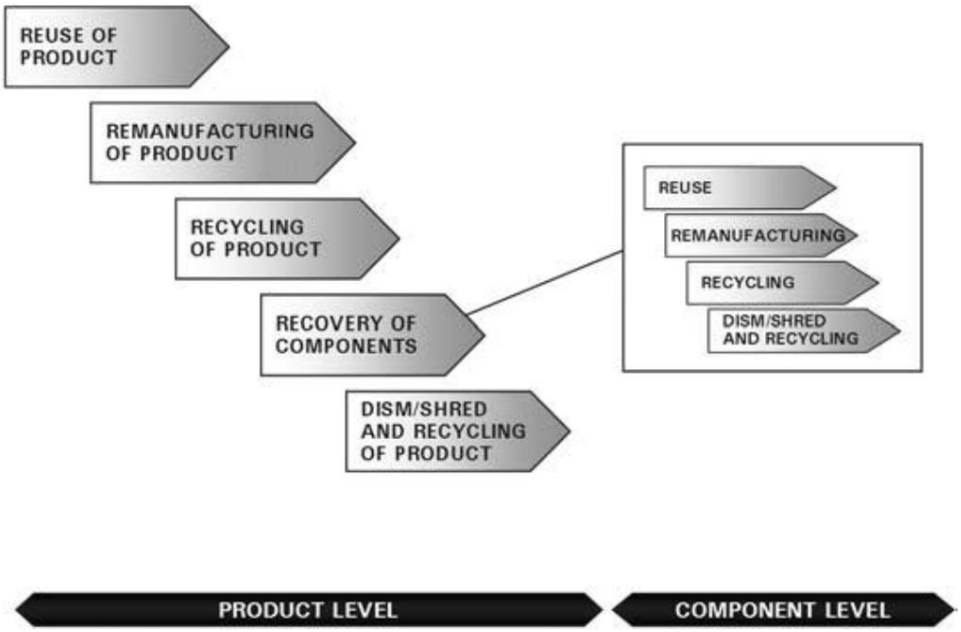


Fig 2.6 End of life Strategies and recovery options

In order to evaluate the effect that design choices can have in terms of disassembling ability and recyclability, certain indices can be used that quantify various characteristics of the constructional system.

 The below model, evidencing the depth of disassembly levels, correlates the successive phases of disassembly with the corresponding parts that become separated and is represented in tree diagram. A graphical representation of this kind is clearly helpful for simplifying the architecture because if the diagram is structured on numerous levels, the product is characterized by great structural depth and consequently by insufficient disassembly.

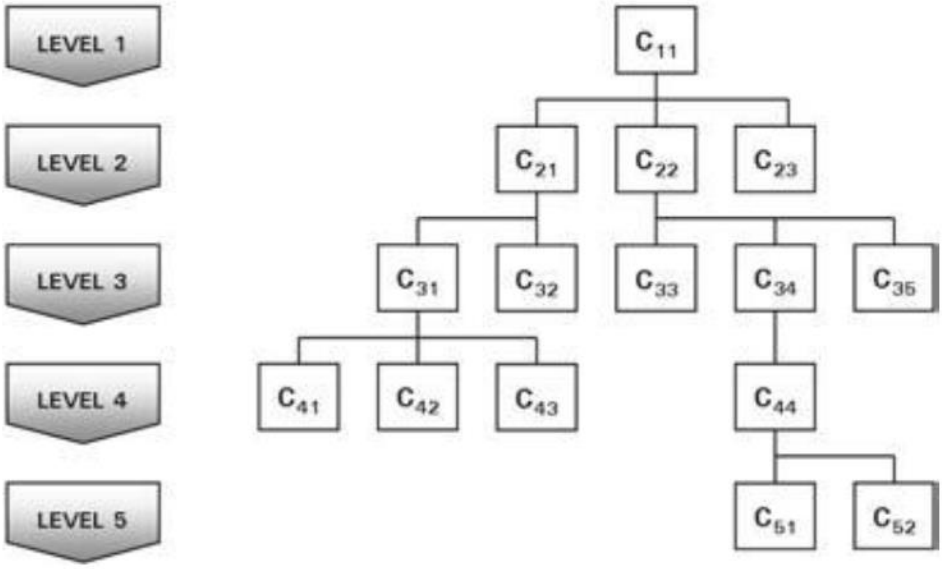


Fig 2.7 Tree diagram for the analysis of disassembly depth

The specific determinant factors for each of end of life strategy are

- Reuse—Physical or mechanical deterioration; risk of damage; technological obsolescence; reliability and durability of components and system; ease of disassembly
- Remanufacturing—Physical or mechanical deterioration; technological obsolescence; reliability and durability of components and system; ease of disassembly and cheapness of remanufacturing processes
- Recycling—Physical or mechanical deterioration; technological obsolescence; reliability and durability of components and system; recyclability and value of materials; ease of material separation and cheapness of recycling processes

HUMAN FACTORS IN PRODUCT DESIGN

Human Factors Engineers strive to produce designs that better meet the capabilities, limitations, and needs of the people who use them. The performance and reliability of any product depends upon the effectiveness of both the product design and the human user

Regardless of the hardware and software performance characteristics of a design, some design results in better human performance and are safer than others

The overall design goal is to ensure effective and safe human performance. Since poor performance and human error are major causes of product failure, understanding and designing to human capacities are critical.

Ergonomics or human engineering started during world war-II, when personnel could not properly operate some of the developed complex systems. It is then called as human factors or ergonomics.

Human factors and ergonomics discovers and applies information about human behavior, abilities, limitations, and other characteristics of the design of tools, machines, systems, tasks, jobs, and environments for productive, safe, comfortable, and effective human use.
In this context, human factors and ergonomics deals with a broad scope of problems relevant to the product design and evaluation of work systems, consumer products, and working environments in which human-machine interactions affect human performance and product usability.
In the field of safety, human factors and ergonomics there is a hazard control hierarchy that is used to reduce the likelihood and severity of injury. This hierarchy consists of an ordered set of approaches or procedures for dealing with potential hazards during the product design.
The key elements in this sequence are:

- Design out the hazard
- Guard against the hazard
- Warn of the hazard

MODELING AND SIMULATION IN PRODUCT

Life Cycle Design consists of a design intervention that incorporates all the phases of the product's life cycle (development, production, use, maintenance and repair, retirement, recovery) in the entire design process, from the phase of concept definition to that of detailed design development.
The traditional evolutionary approach is based on the feedback of information flows used to improve the design intervention; a powerful alternative can consist of already predicting the consequences of design choices on the product's life cycle during the solution synthesis phase itself. This is equivalent to simulating the life cycle of the product in the early phases of the design process.
To correlate the set of design choices and product performance, the methodological approach proposed here is based on advanced Failure Mode and Effect Analysis (FMEA), scheduling the development of a product behavior model based on its function rather than on its structure.
This is based on the definition of the main elementary functionalities of the constructional system, on the set of determining variables required for the behavior under examination to take place (initial state) and final conditions reached after the function has taken place (final state), and of the performance characteristics regulating the behavior, which can generally be expressed using mathematical models.
Taking into consideration a defined set of materials characterized by decay curves, it is possible to evaluate the indices that provide information on the duration of components (Index of Duration—DI) and on safety (Dynamic Criticality Factor —DCF) over the life cycle.
On these bases, the simulation of the system's functionalities allows the identification of possible failures and their classification in terms of their effect on components. It is then possible to derive the criticality of the system through analysis of the configuration of the model (blocks in a series or parallel), the danger of a failure, and the residual performance level.
This information directly influences possible strategies for improving the environmental performance of the life cycle and optimization of the useful life (through servicing operations on the system) and the recovery of resources at the end of life. In this regards reference is made to the calculation models already available for both the evaluation of the economic impact of the servicing systems during their useful life.

The simulation procedure follows the development of models proposed and is summarized in below figure 2.8. It is based on direct relation between the design parameters that constitute the variables to be optimized, and the objective functions that quantify the product's performance in the main phases of the life cycle.
This relation is structured using behavior model of the system and is expressed analytically using the indicators of component duration and failure derived from the deterioration of performance over time.

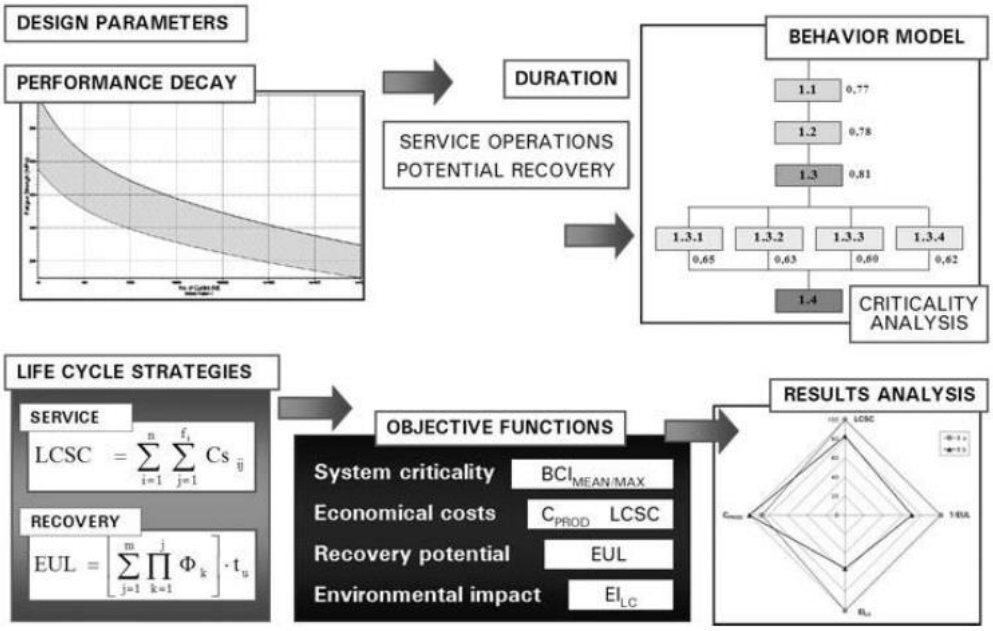


Fig 2.8 Methodological framework and tools

Analysis of any product is basically considered on its performance and its criticality. A case study representation is as follows in fig 2.9

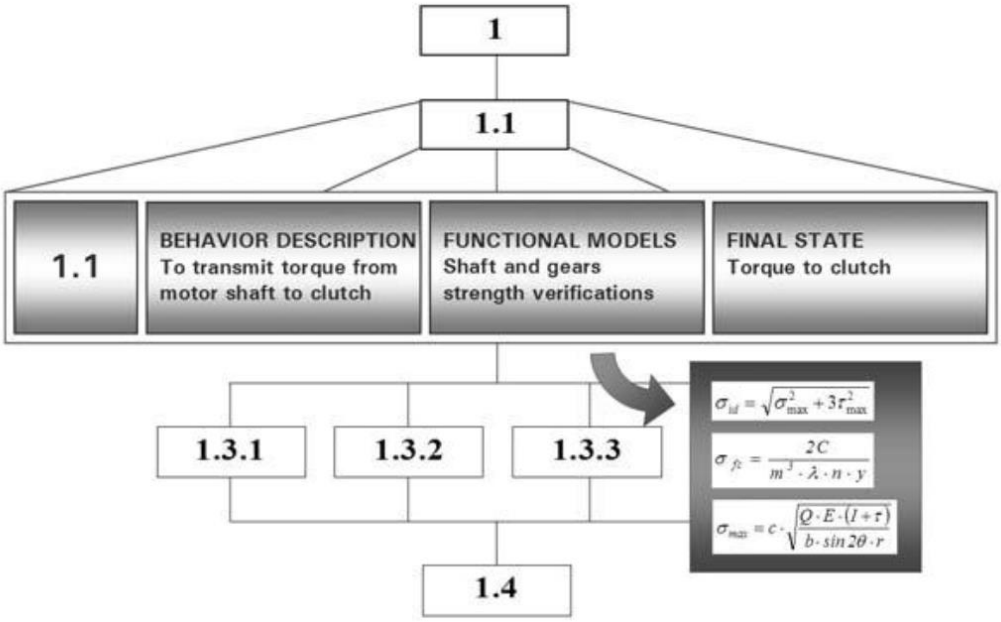


Fig 2.9 Model of system behavior – a case study

At the first, analysis of the performance of the product is assessed in the product life cycle and the high number of components failure and substitutions are identified. The analysis will thus be extended into the end of life phase where the little possibility of component reuse is

identified. Finally the environmental impact of the entire life cycle of the product is analyzed and the effectiveness of product in the market can be forecasted.

Question Bank:

1. What is Engineering design? Explain.
2. Sketch and explain the product design process.
3. Sketch and explain the recovery strategies at end of life.
4. Explain design process models and reference models.
5. Explain with a graph strategy for recovery at end of life.
6. Explain the Concurrent Engineering.
7. Sketch and explain the comparison between sequential and concurrent engineering in product development process.
8. List and explain the most common DFX techniques used in the product design and development process.
9. Explain the design for 'X' and design centred model.
10. With a neat sketch explain engineering design process.
11. Explain the organization and decomposition in product design.
12. Discuss the various steps involved in product design.
13. Describe the recycling and Human factors in product design.
14. Explain the human factors to be considered in product design.
15. Write a short note on modelling and simulation of a product.